

Replicating the Saving Glut of the Rich

Independent reconstruction of selected results in Mian, Straub, and Sufi (July 2025)

Method and data

The empirical problem is to move from observed holders to *ultimate* owners. For direct ownership matrix $\bar{M}_t$ —rows are issuers and columns are holders—the revised paper sums every intermediary path. With Q_t the intermediary block and B_t the direct exits to terminal owners, our Python implementation solves

$$\Omega_t = (I - Q_t)^{-1} B_t, \quad \rho(Q_t) < 1.$$

Financial Accounts/FWTW matrices identify bilateral stocks; DINA allocates the household sector across wealth groups; NIPA closes aggregate income and saving; and asset returns separate active saving from valuation changes, $\Theta_{g,t} = \sum_a (W_{g,t}^a - W_{g,t-1}^a - \pi_t^a W_{g,t-1}^a)$.

We first ran the complete February 2021 Stata master under StataNow SE 19.5. After portability-only path changes, dependency bootstrapping, and suppressing one interactive browser call, it completed and regenerated the major outputs for the paper version it accompanies. That package generally ends in 2016 and uses seven stage-specific unveiling rounds. The 2025 paper instead reports 34 instruments, 23 intermediaries plus four ultimate owners, a revised completion, a full matrix/Leontief solve, and data through 2019. We therefore reimplemented the 2025 definitions in Python from the earliest usable supplied inputs and label every remaining version boundary.

Replication results

Exhibit	Evidence layer	Result	Binding limitation
Figure 1	Old-input reconstruction	Close level match: $r = 0.995$ and MAE = 5.3 pp over 1963–2016.	Exact 2025 completed matrices and indirect-share denominator are unavailable.
Figure 5	Old-input reconstruction	High fidelity: top 1% / bottom 99% correlations 0.992 / 0.997; MAE 0.40 / 0.42 pp.	Same saving transformation, but the authors’ supplied inputs stop in 2016.
Figure 8	Same-vintage method reconstruction	Trend and borrowing match: $r = 0.994/0.995$ and MAE = 2.33/1.31 pp for top 1% / bottom 99%.	The kit saves 351 direct fields, not the full 34-instrument by 27-sector cube; coarsening, signed-cell clipping, and an unassigned owner remain.

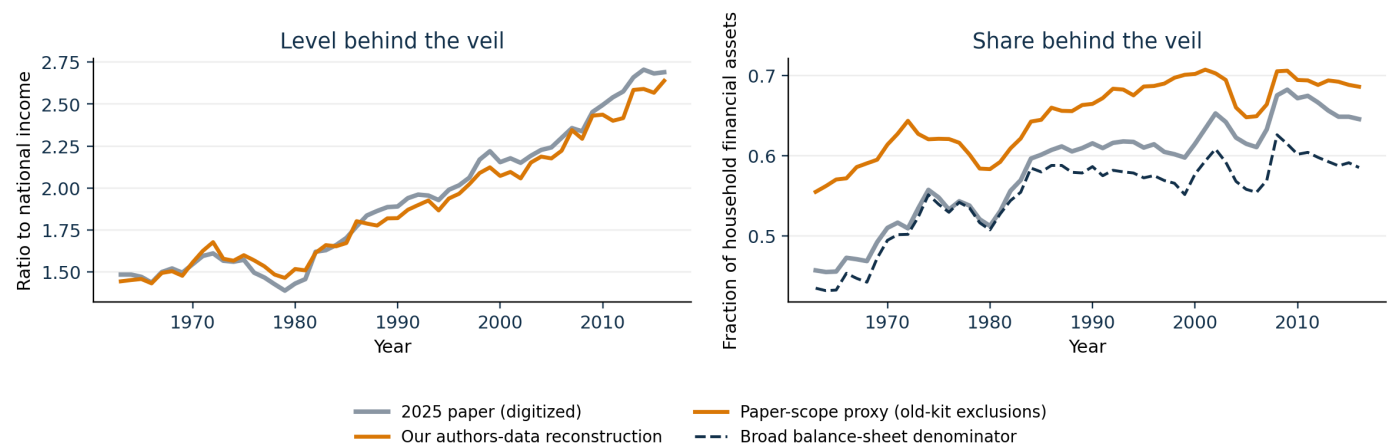
Interpretation and reproducibility boundary

These are version-bounded reconstructions, not exact numerical reproductions. Figures 1 and 5 recover the main 2025 patterns from older author-supplied inputs. For Figure 8, restarting from saved direct cells and applying the full inverse yields 2007 top-1 lending of 11.47 pp, versus 18.02 pp in the paper, while bottom-99 debt is -41.03 pp versus -38.76 pp. Holding the 2021 vintage as fixed as the saved fields permit, replacing seven rounds with the full inverse changes the two series by only 0.26 and 0.75 pp on average. The remaining gap is therefore mainly a data-vintage, completion, taxonomy, and allocation boundary rather than finite-round truncation. A current-public-FWTW rerun is a separate robustness exercise, not an exact recovery of the authors’ values.

All empirical series, comparisons, and displayed metrics are generated by code. The principal entry points are `build_figure1_authors_data.py`, `build_figure5_authors_data.py`, and `build_figure8_2021_direct.py`; tests cover matrix conservation, group partitions, valuation/saving closure, normalization, and debt signs.

Figure 1 — Indirect household ownership

Figure 1: authors-data level match, denominator remains version-specific

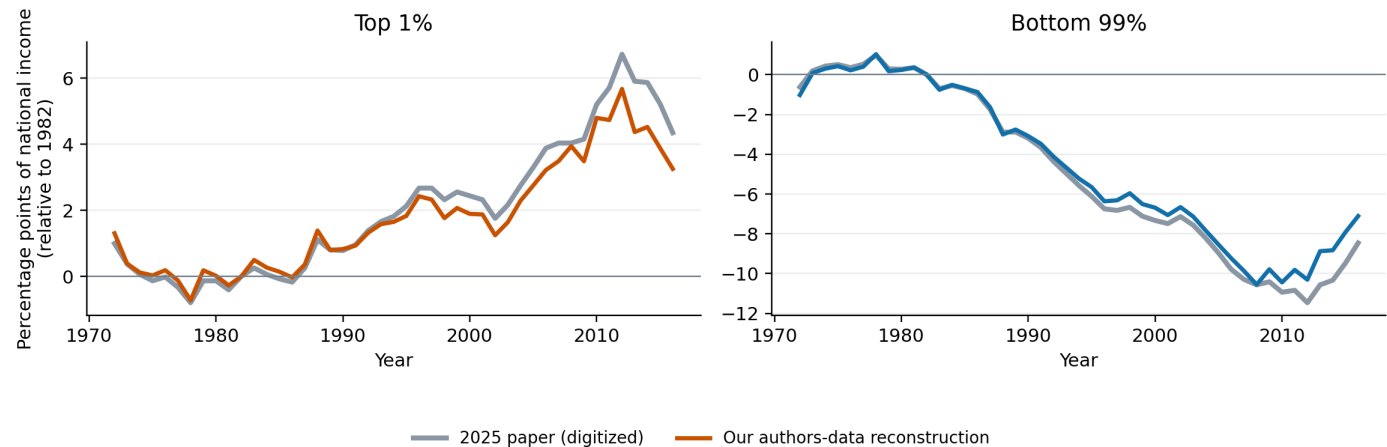


Finding. The old-input stock reconstruction follows the revised Figure 1 level closely. The plotted share remains an explicitly labelled old-kit denominator proxy.

Source: generated by Code/scripts/build_figure1_authors_data.py; paper panel digitized from the July 2025 PDF.

Figure 5 — Saving by wealth group

Figure 5: same transformation, older data stop in 2016

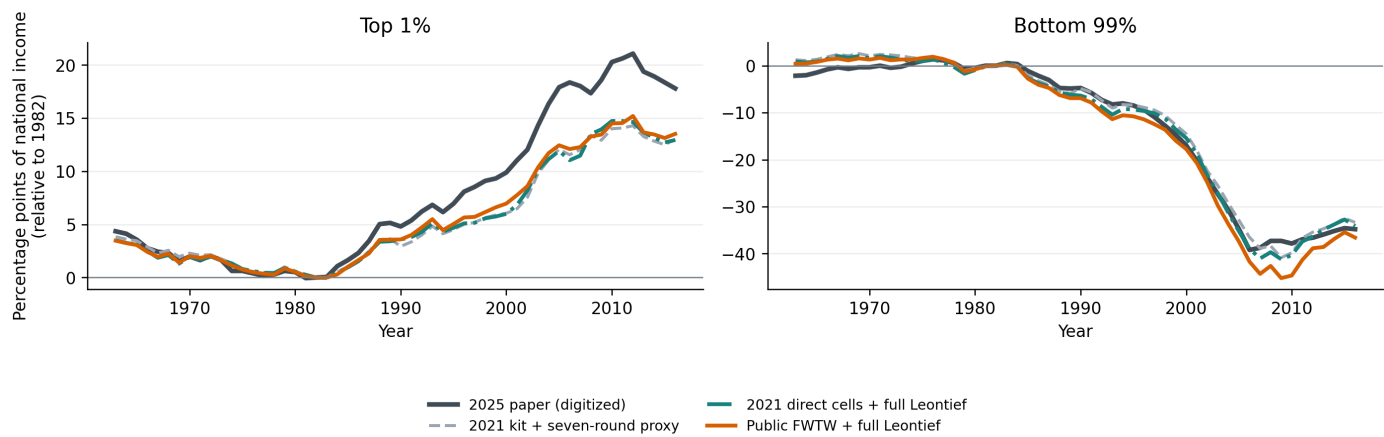


Finding. The 2025 saving transformation applied to 2021-kit inputs reproduces the divergence between top-1 and bottom-99 saving through 2016.

Source: generated by Code/scripts/build_figure5_authors_data.py; paper curves digitized from the July 2025 PDF.

Figure 8 — Net household debt by wealth group

Figure 8: separating the operator experiment from the public-data rerun



Finding. All three routes reproduce the post-1982 split. The primary same-vintage route restarts before unveiling and applies the full inverse; the seven-round result is its lineage benchmark and the public-FWTW route is a mixed-vintage robustness check. The small same-vintage operator gap shows that missing 2025 matrix coverage, rather than truncation alone, drives the remaining top-1 level difference.

Source: generated by Code/scripts/build_figure8_method_comparison.py; paper curves digitized from the July 2025 PDF.